```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("heart.csv")
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Next steps: Generate code with df New interactive sheet

```
df.shape
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    int64
 1   sex       303 non-null    int64
 2   cp        303 non-null    int64
 3   trestbps  303 non-null    int64
 4   chol      303 non-null    int64
 5   fbs       303 non-null    int64
 6   restecg   303 non-null    int64
 7   thalach   303 non-null    int64
 8   exang     303 non-null    int64
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64
 11  ca        303 non-null    int64
 12  thal      303 non-null    int64
 13  target    303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

```
df.shape
```

```
(303, 14)
```

```
df.dtypes
```

	0
age	int64
sex	int64
cp	int64
trestbps	int64
chol	int64
fbs	int64
restecg	int64
thalach	int64
exang	int64
oldpeak	float64
slope	int64
ca	int64
thal	int64
target	int64

dtype: object

```
df.isnull().sum()
```

	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

```
df.duplicated().sum()
df.drop_duplicates(inplace=True)
```

```
df.isnull().sum()
```

	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

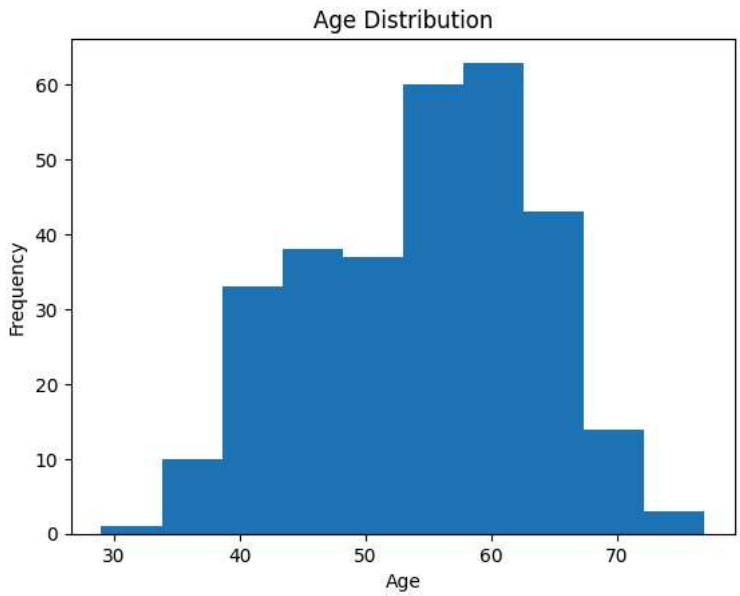
```
# Age stats
df['age'].mean()
df['age'].median()
df['age'].mode()
df['age'].std()
df['age'].var()
df['age'].max() - df['age'].min()
(df['age'].max() + df['age'].min())/2
```

```
53.0
```

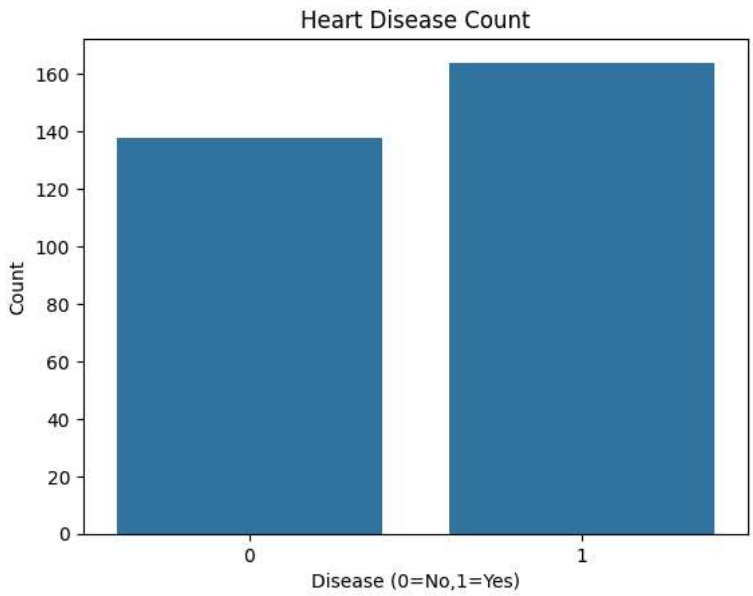
```
# Cholesterol stats
df['chol'].mean()
df['chol'].median()
df['chol'].mode()
df['chol'].std()
df['chol'].var()
df['chol'].max() - df['chol'].min()
(df['chol'].max() + df['chol'].min())/2
```

```
345.0
```

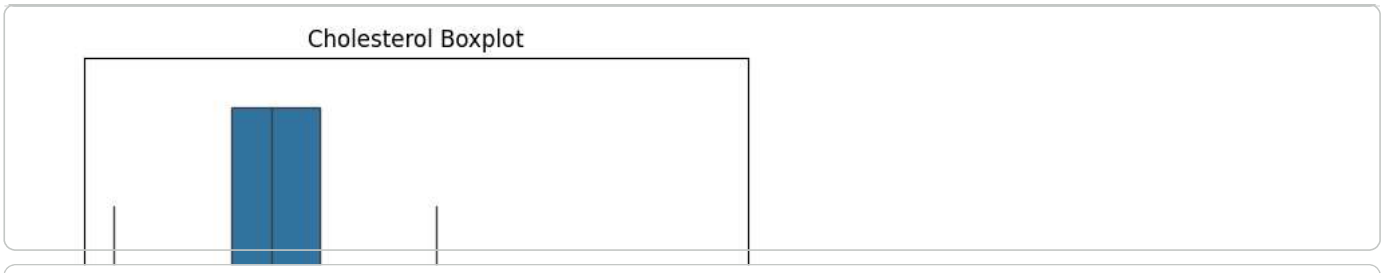
```
plt.hist(df['age'])
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

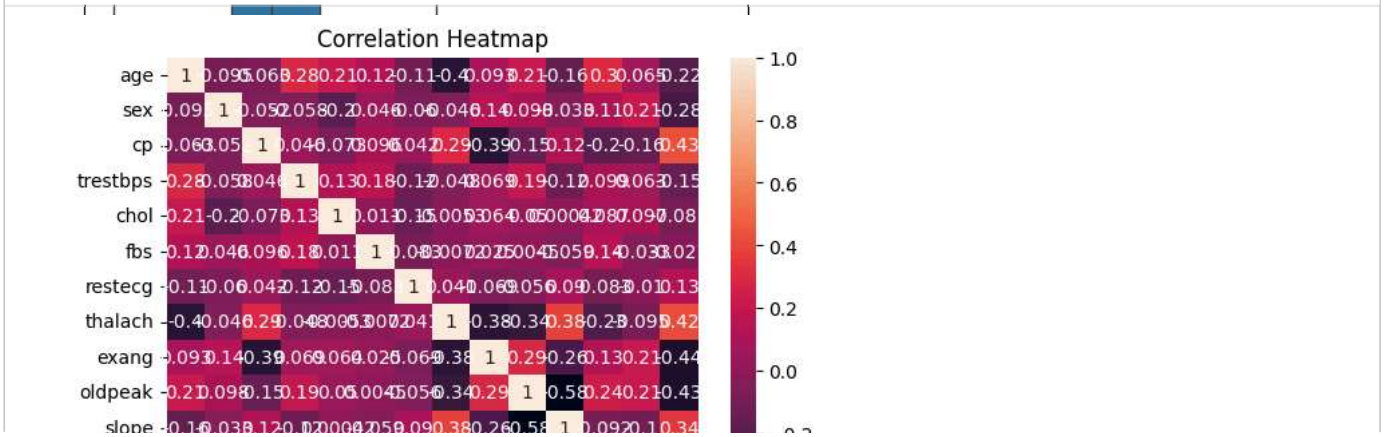



```
sns.countplot(x='target', data=df)
plt.title("Heart Disease Count")
plt.xlabel("Disease (0=No,1=Yes)")
plt.ylabel("Count")
plt.show()
```




```
sns.boxplot(x=df['chol'])
plt.title("Cholesterol Boxplot")
plt.xlabel("Cholesterol")
plt.show()
```




```
corr = df.corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```